Lecture 6 - Support Vector Machine (SVM)

ECE 447: Data Analysis and Machine Learning for Engineers

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Capabilities of An First/Second Year Undergrad in Research

Sparse Autoencoder Neural Operators: Model Recovery in Function Spaces

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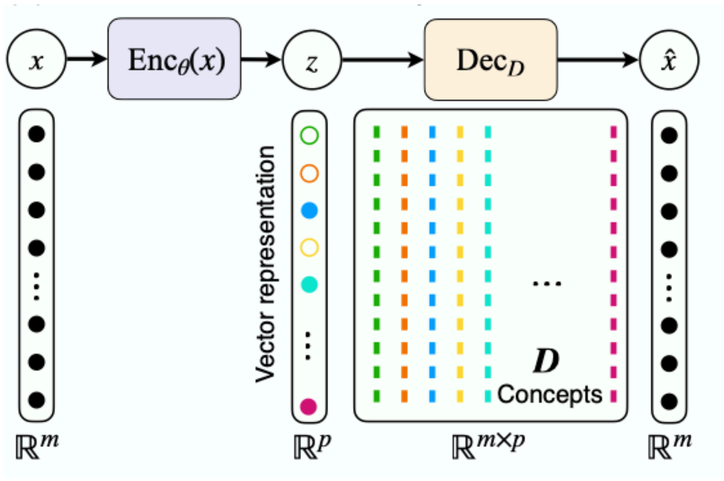
presented at UniReps workshop at NeurIPS 2025

Mechanistic Interpretability with Sparse Autoencoder Neural Operators



Ailsa, Anima and I submitted another paper last night to ICML.

This Lecture Covers ...

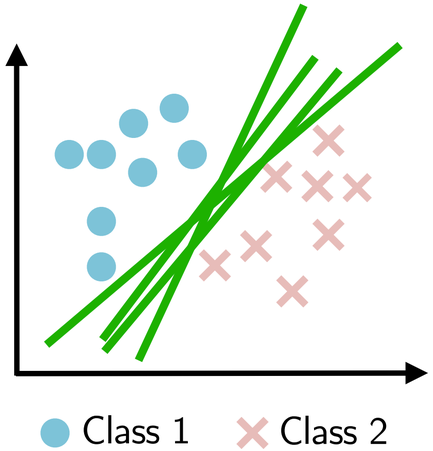
- Support vector machines (SVMs): separable case
- Support vector machines (SVMs): non-separable case
- Optimization perspective behind SVMs

Problem Setup

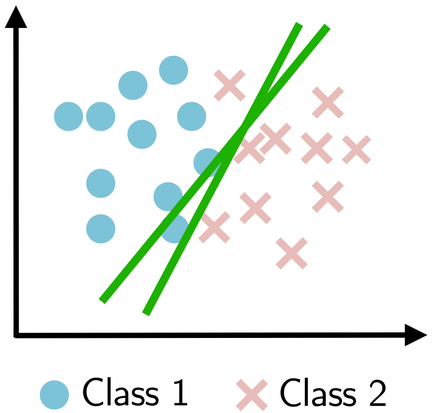
***Which one** of these lines we want as the classification boundary?*

Given a set of data points $\{\boldsymbol{x}_i \in \mathbb{R}^p, y_i \in \{\overset{c_1}{+1}, \overset{c_2}{-1}\}\}_{i=1}^n$, for $i = 1, \ldots, n$

Linearly separable



Linearly non-separable



Goal: Find a linear function $f(\boldsymbol{x}) = \boldsymbol{\theta}_1^\top \boldsymbol{x} + \theta_0$ that gives a class assignment $\hat{c}(\boldsymbol{x}_i)$

$$\begin{cases} \hat{c}(\boldsymbol{x}_i) = +1, \text{ if } f(\boldsymbol{x}_i) \geq b \\ \hat{c}(\boldsymbol{x}_i) = -1, \text{ if } f(\boldsymbol{x}_i) \leq -b \end{cases} \quad \text{for some } b$$

Support Vector Machine (SVD): An Optimization Formulation

There exists **infinitely** many $f(\boldsymbol{x})$ and b that satisfy $\hat{c}(\boldsymbol{x}_i) = c(\boldsymbol{x}_i)$ to perfectly separate the classes.

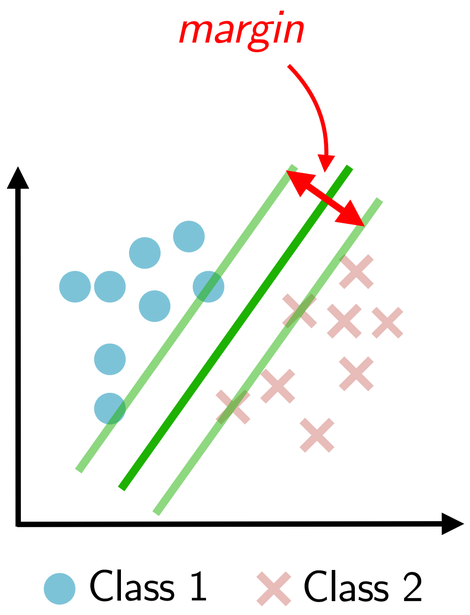
Support Vector Machine (SVM)

(separability condition)

I) Choose the line that perfectly **separates** the classes.

(maximum margin classifier)

II) Choose the line that maximizes the **margin** between the classes.



Formulate Separability Condition for SVM (Separable Case)

Suppose $f(\boldsymbol{x})$ separates the data classes. Then,

$$\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0 \geq b \quad \forall\ c(\boldsymbol{x}_i) = 1$$

Multiply the inequalities on both side by y_i, to get the *separability condition*

$$y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) \geq b$$

Suppose $\boldsymbol{z} \in \mathbb{R}^p$ is such that $\boldsymbol{\theta}_1^\top \boldsymbol{z} + \theta_0 = b$. Then,

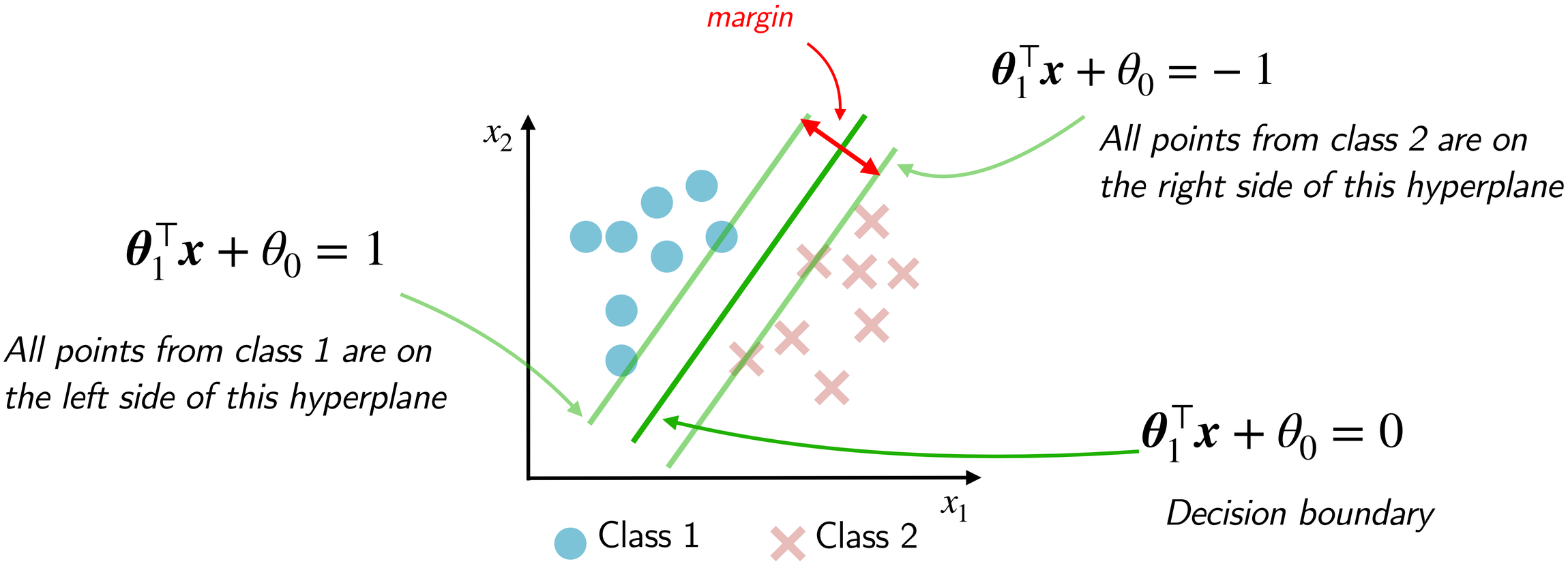
$$\tilde{\boldsymbol{\theta}}_1 = \frac{\boldsymbol{\theta}_1}{\boldsymbol{\theta}_1^\top \boldsymbol{z} + \theta_0} \qquad \tilde{\theta}_0 = \frac{\theta_0}{\boldsymbol{\theta}_1^\top \boldsymbol{z} + \theta_0}$$

$$y_i(\tilde{\boldsymbol{\theta}}_1^\top \boldsymbol{z} + \tilde{\theta}_0) \geq 1 \quad \Rightarrow$$

$$\boxed{y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) \geq 1}$$

separability condition

SVM Separability Condition Visualization (Separable Case)

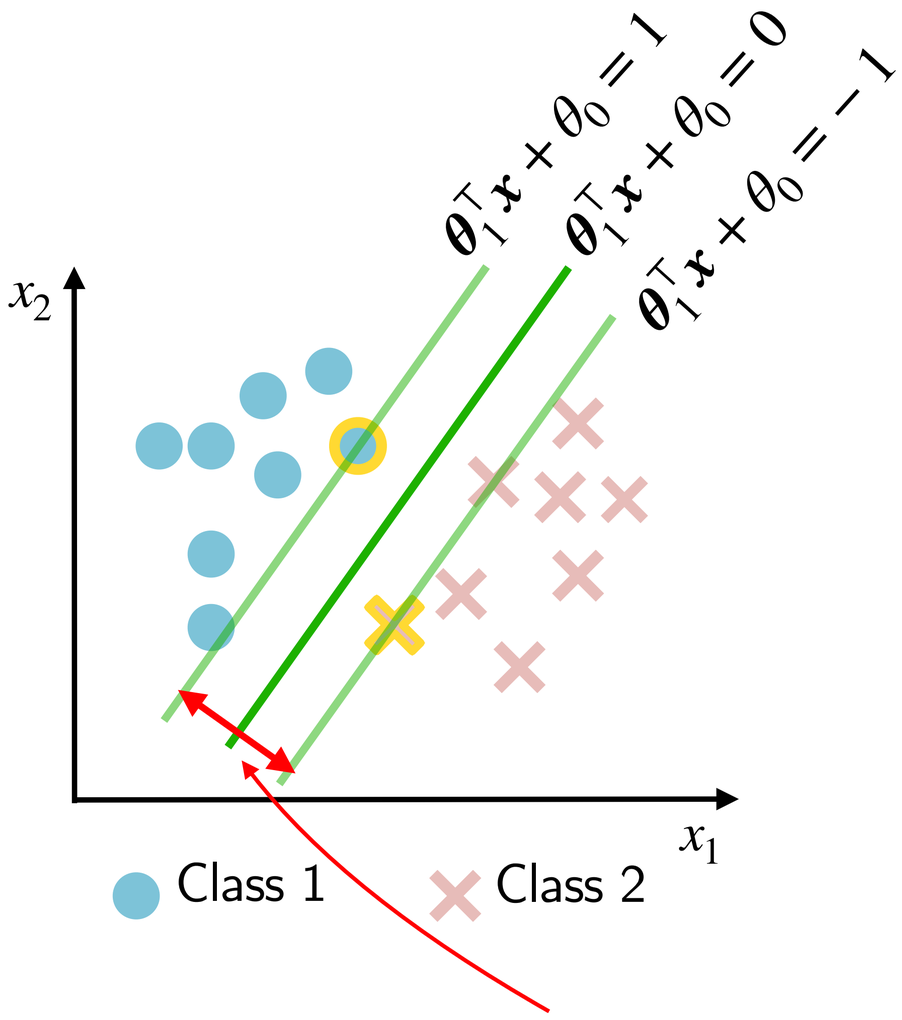


SVM Margin Visualization

To find a classifier that maximize the margin,

$$\exists\ \boldsymbol{x}_i^+ \text{ s.t. } c(\boldsymbol{x}_i^+) = +1 \quad \text{and} \quad \boldsymbol{\theta}_1^\top \boldsymbol{x}_i^+ + \theta_0 = +1$$

$$\exists\ \boldsymbol{x}_i^- \text{ s.t. } c(\boldsymbol{x}_i^-) = -1 \quad \text{and} \quad \boldsymbol{\theta}_1^\top \boldsymbol{x}_i^- + \theta_0 = -1$$



Intuition: The support hyperplane$^+$ touch one example from c_1 and hyperplane$^-$ touch one example from c_2.

How to quantify the margin?

Why? Otherwise, there will still be room to increase (maximize) the margin (distance between the two hyperplanes $\{\boldsymbol{\theta}_1^\top \boldsymbol{x} + \theta_0 = 1\}$ and $\{\boldsymbol{\theta}_1^\top \boldsymbol{x} + \theta_0 = -1\}$).

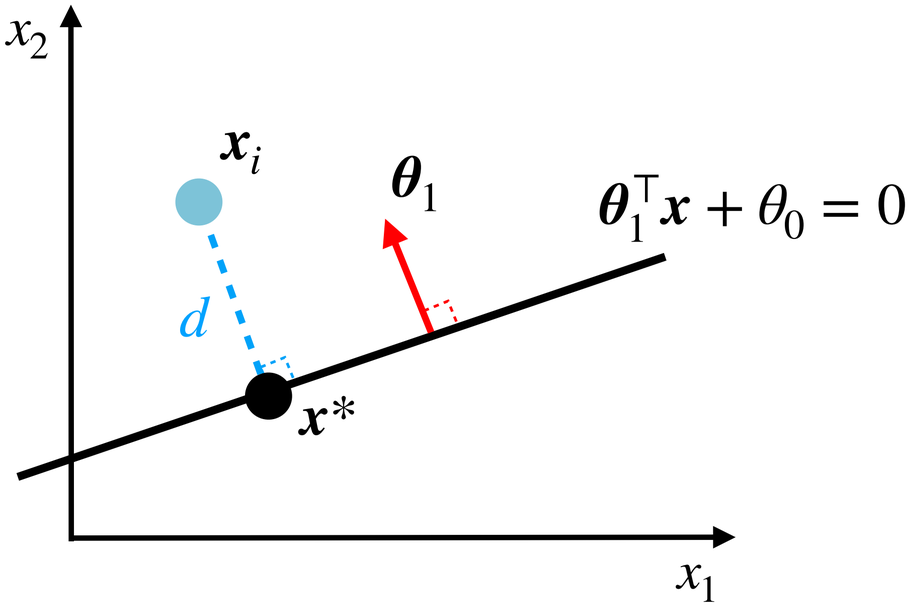
Distance from a Point to a Hyperplane (I)

Intuition: $\boldsymbol{x}_i - \boldsymbol{x}^*$ is along the direction of $\boldsymbol{\theta}_1$.

Given the hyperplane $\mathrm{H} = \{\boldsymbol{\theta}_1^\top \boldsymbol{x} + \theta_0 = 0\}$,

the point, on the hyperplane, closest to $\boldsymbol{x}_i$ is

$$\boldsymbol{x}^* = \arg\min_{\boldsymbol{x} \in \mathrm{H}} \|\boldsymbol{x}_i - \boldsymbol{x}\|_2$$



If $\boldsymbol{x}_A, \boldsymbol{x}_B \in \mathrm{H}$, then $\langle \boldsymbol{x}_i - \boldsymbol{x}^*, \boldsymbol{x}_A - \boldsymbol{x}_B \rangle = 0$, i.e.,

$$\exists\ \beta \text{ s.t. } \boldsymbol{x}_i - \boldsymbol{x}^* = \beta \boldsymbol{\theta}_1$$

$$\boldsymbol{x}_i - \boldsymbol{x}^* = \beta \boldsymbol{\theta}_1 \quad \Rightarrow \quad \boxed{d := \|\boldsymbol{x}_i - \boldsymbol{x}^*\| = \|\beta \boldsymbol{\theta}_1\|}$$

Continue next slide to derive β and margin.

Distance from a Point to a Hyperplane (II)

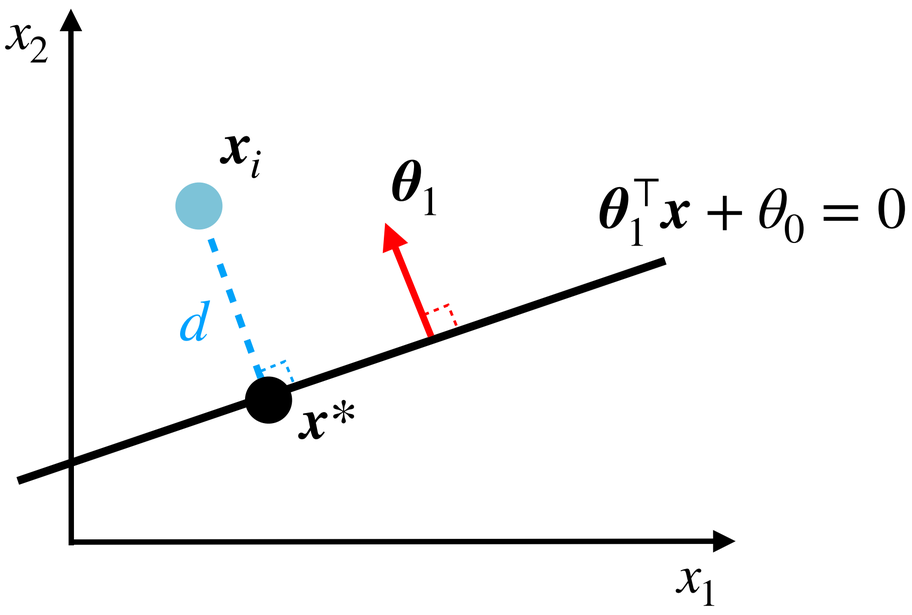
Given $\boldsymbol{x}_i - \boldsymbol{x}^* = \beta \boldsymbol{\theta}_1$, we find β as following

$$\boldsymbol{\theta}_1^\top (\boldsymbol{x}_i - \boldsymbol{x}^*) = \beta \|\boldsymbol{\theta}_1\|_2^2$$

Denote $\theta_0 := -\boldsymbol{\theta}_1^\top \boldsymbol{x}^*$. Then,

$$\beta = \frac{\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0}{\|\boldsymbol{\theta}_1\|_2^2}$$

$$d = \|\beta \boldsymbol{\theta}_1\| = \frac{|\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0|}{\|\boldsymbol{\theta}_1\|_2}$$

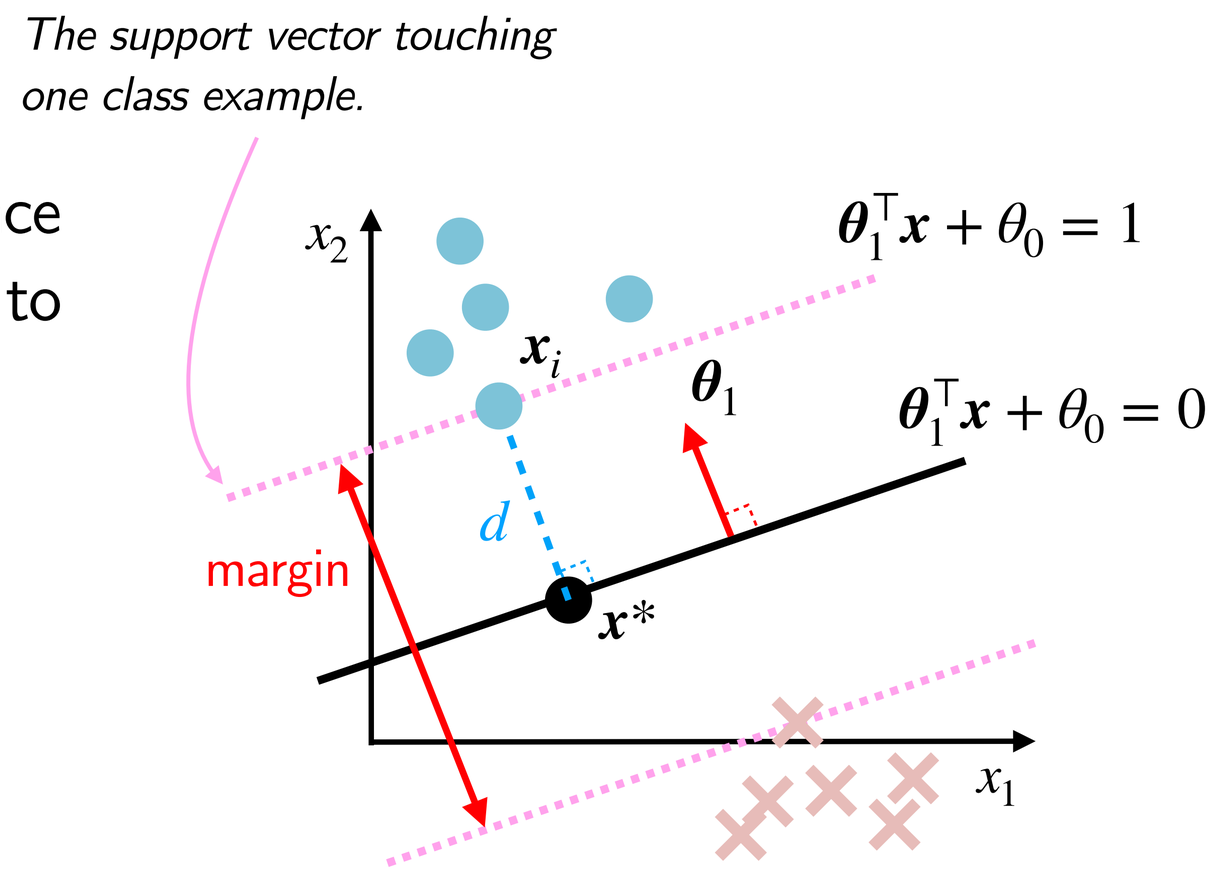


Margin of SVM

We are interested in quantifying the distance from a point that is on the support vector to the decision boundary.

For $\boldsymbol{x}_i \in \{\boldsymbol{\theta}_1^\top \boldsymbol{x} + \theta_0 = 1\}$,

$$d = \frac{1}{\|\boldsymbol{\theta}_1\|_2}$$



The margin is twice as much the above distance. $\Rightarrow$ $\boxed{\text{margin} := \frac{2}{\|\boldsymbol{\theta}_1\|_2}}$

Optimization Formulation for SVM (Separable Case)

We are now ready to write down the optimization

Combining the separability condition with maximizing the margin, we get

$$\max_{\boldsymbol{\theta}} \frac{2}{\|\boldsymbol{\theta}_1\|_2} \quad \text{s.t.} \quad y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) \geq 1 \qquad \text{for } i = 1,\ldots,n$$

$$\min_{\boldsymbol{\theta}} \frac{\|\boldsymbol{\theta}_1\|_2}{2} \quad \text{s.t.} \quad y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) \geq 1 \qquad \text{for } i = 1,\ldots,n$$

$$\boxed{\min_{\boldsymbol{\theta}} \frac{1}{2}\|\boldsymbol{\theta}_1\|_2^2 \quad \text{s.t.} \quad y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) \geq 1 \qquad \text{for } i = 1,\ldots,n}$$

Lagrangian and Optimality Conditions for SVM (Separable Case)

Lagrangian of the objective is

$$L(\boldsymbol{\theta}_1, \boldsymbol{\theta}_0, \boldsymbol{\lambda}) = \frac{1}{2}\|\boldsymbol{\theta}\|_2^2 \; - \; \sum_{i=1}^{n} \lambda_i \left(y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) - 1 \right), \quad \boldsymbol{\lambda} \in \mathbb{R}^n$$

The optimality conditions are

(I) $\quad \nabla_{\boldsymbol{\theta}_1} L(\boldsymbol{\theta}_1, \boldsymbol{\theta}_0, \boldsymbol{\lambda}) = 0 \quad \Leftrightarrow \quad \hat{\boldsymbol{\theta}}_1 = \sum_{i=1}^{n} \lambda_i y_i \boldsymbol{x}_i$

(II) $\quad \nabla_{\theta_0} L(\boldsymbol{\theta}_1, \boldsymbol{\theta}_0, \boldsymbol{\lambda}) = 0 \quad \Leftrightarrow \quad \sum_{i=1}^{n} \lambda_i y_i = 0$

(III) $\quad \begin{cases} \lambda_i \left(y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) - 1 \right) = 0 & \text{for } i = 1, \ldots, n \\ \lambda_i \geq 0 \end{cases}$

If $\lambda_i \neq 0$, *then* $y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) = 1$

Unconstraint Optimization for SVM (Separable Case)

The dual problem

The optimization now can be written as a function of *the Lagrange multiplier*, $\boldsymbol{\lambda}$

Coming from ***(III)***

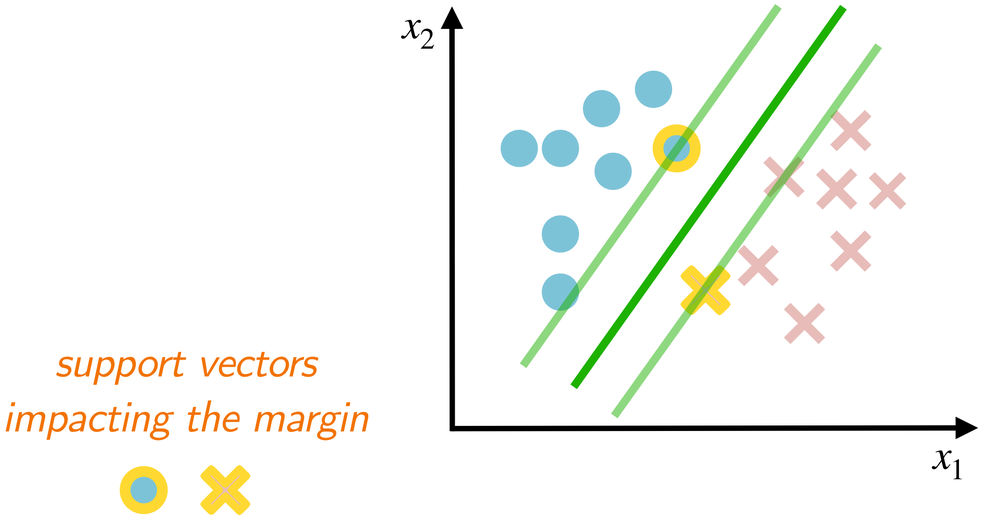
Coming from $\|\boldsymbol{\theta}_1\|_2^2$

$$\max_{\lambda} \quad \sum_{i=1}^{n} \lambda_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j}^{n} \lambda_i \lambda_j \, y_i y_j \, \boldsymbol{x}_i^\top \boldsymbol{x}_j$$

$$\text{s.t.} \quad \lambda_i \geq 0, \qquad \text{for } i = 1, \ldots, n$$

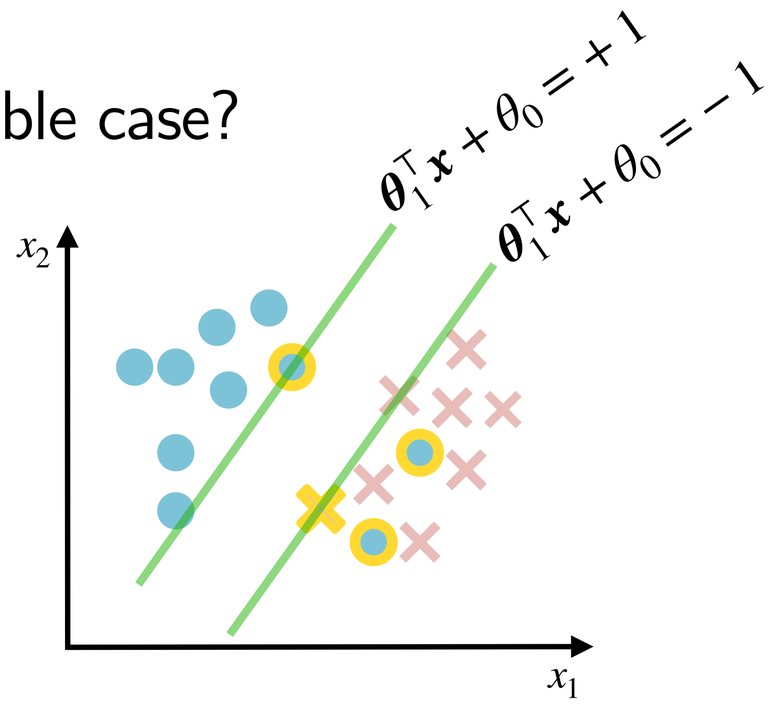
$$\sum_{i=1}^{n} \lambda_i y_i = 0$$

Remarks: from **(III)**, the *only* examples with $\lambda_i \neq 0$ are those on the support hyperplanes.



The Non-Separable Linear Classification

What is the role of misclassified examples for the non-separable case?



Zero-One Class

$$\mathscr{L}(y, f(\boldsymbol{x})) = \begin{cases} 1, & yf(\boldsymbol{x}) \leq 0 \\ 0, & \text{otherwise} \end{cases}$$

Separable case:

(100% classification accuracy)

$$\mathscr{L}(y, f(\boldsymbol{x})) = 0 \quad \forall i = 1, \ldots, n$$

$$y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) \geq 0 \quad \forall i = 1, \ldots, n$$

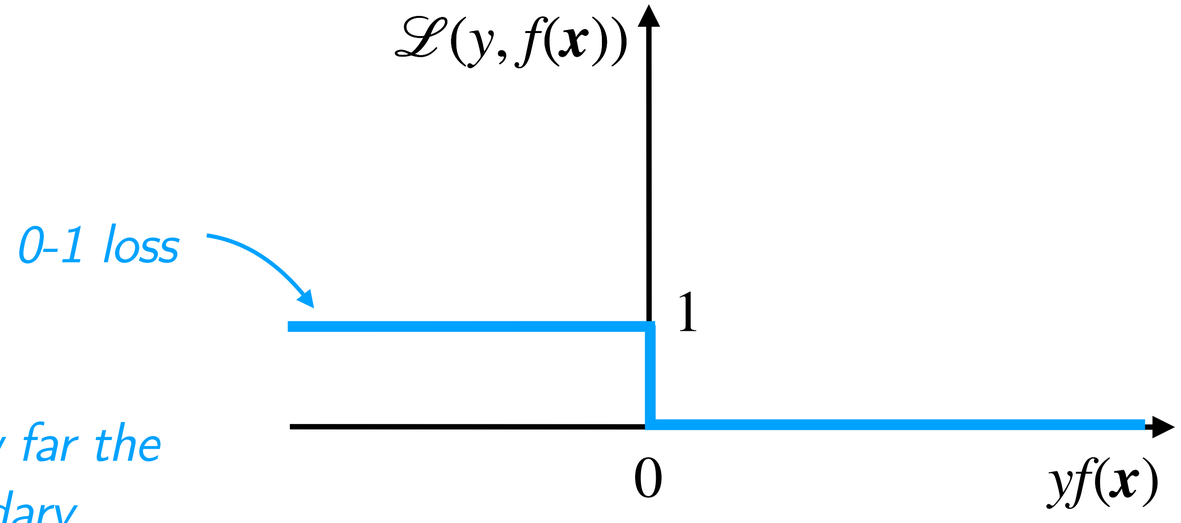
Zero-One Loss for Non-Separable Linear Classification

Non-separable case:

$$\exists j \in \{1, \ldots, n\} \quad \text{s.t.} \quad y_j(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) < 0$$

Zero-one loss penalizes the miss-classified example by loss value of 1.

$$\mathscr{L}(y_j, f(\boldsymbol{x}_j)) = 1$$



Intuition: in zero-one loss, it does not matter how far the miss-classified example is from the decision boundary.

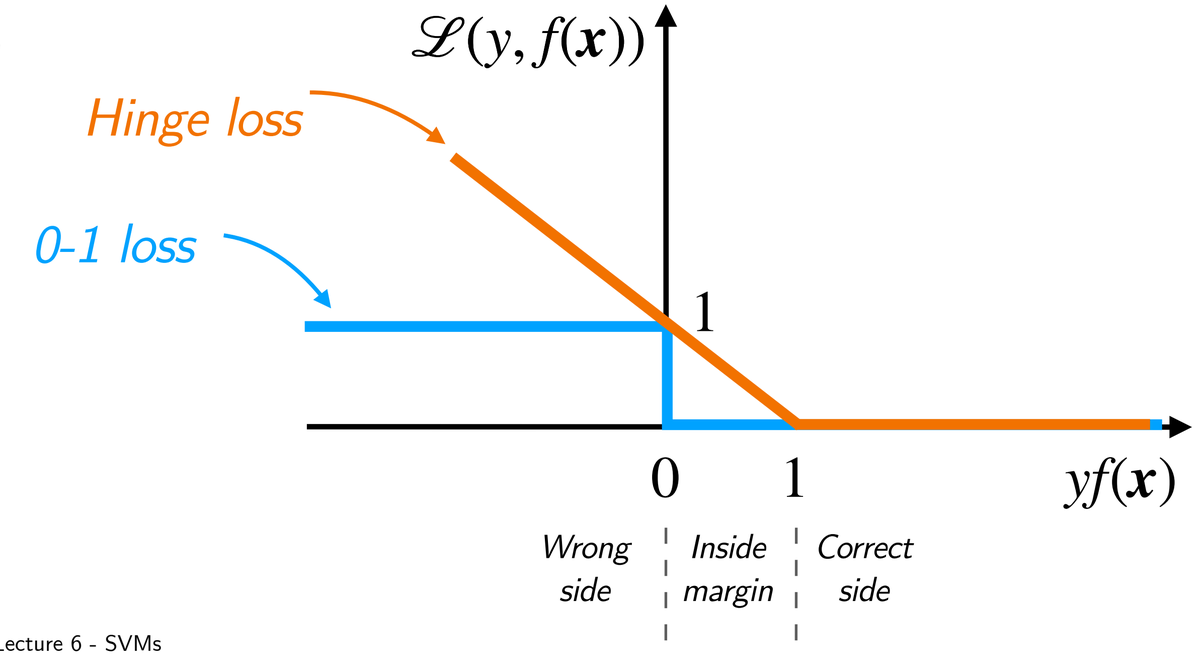
Hinge Loss for Non-Separable SVM

Hinge Loss

$$\mathscr{L}_H(y, f(\boldsymbol{x})) \;=\; \max(\; 0 \;,\; 1 - yf(\boldsymbol{x}) \;)$$

We define a cost $\zeta > 0$ to penalize points inside the margin or on the wrong side.

$$1 - y(\boldsymbol{\theta}_1^\top \boldsymbol{x} + \theta_0) \leq \zeta$$



SVM Optimization for Non-Separable Case

Separable case

$$\min_{\boldsymbol{\theta}} \ \frac{1}{2}\|\boldsymbol{\theta}_1\|_2^2 \text{ s.t. } y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) \geq 1 \quad \text{for } i = 1,\ldots,n$$

Non-separable case:

$$\min_{\boldsymbol{\theta}} \ \frac{1}{2}\|\boldsymbol{\theta}_1\|_2^2 \ + \ C\sum_{i=1}^{n} \zeta_i$$

$$\text{s.t.} \quad y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) \geq 1 - \zeta_i, \quad \text{for } i = 1,\ldots,n$$

$$\zeta_i \geq 0, \quad \text{for } i = 1,\ldots,n$$

Correct side: $-a \leq 0 \leq \zeta_i, \quad a > 0$

On margin: $0 \leq \zeta_i$

Inside margin or wrong side: $a \leq \zeta_i, \quad a > 0$

The Lagrangian for Non-Separable SVM

$$L(\boldsymbol{\theta}_1, \boldsymbol{\theta}_0, \boldsymbol{\zeta}, \boldsymbol{\lambda}, \boldsymbol{\gamma}) = \frac{1}{2}\|\boldsymbol{\theta}_1\|_2^2 + C\sum_{i=1}^{n}\zeta_i - \sum_{i=1}^{n}\gamma_i\zeta_i - \sum_{i=1}^{n}\lambda_i(y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) - 1 + \zeta_i)$$

The optimality conditions are

(I) $\nabla_{\boldsymbol{\theta}_1} L(\boldsymbol{\theta}_1, \boldsymbol{\theta}_0, \boldsymbol{\zeta}, \boldsymbol{\lambda}, \boldsymbol{\gamma}) = 0 \quad \Leftrightarrow \quad \hat{\boldsymbol{\theta}}_1 = \sum_{i=1}^{n} \lambda_i y_i \boldsymbol{x}_i$

(II) $\nabla_{\theta_0} L(\boldsymbol{\theta}_1, \boldsymbol{\theta}_0, \boldsymbol{\zeta}, \boldsymbol{\lambda}, \boldsymbol{\gamma}) = 0 \quad \Leftrightarrow \quad \sum_{i=1}^{n} \lambda_i y_i = 0$

(III) $\nabla_{\zeta_i} L(\boldsymbol{\theta}_1, \boldsymbol{\theta}_0, \boldsymbol{\zeta}, \boldsymbol{\lambda}, \boldsymbol{\gamma}) = 0 \quad \Leftrightarrow \quad C - \lambda_i - \gamma_i = 0$

(IV) $\begin{cases} \lambda_i \left(y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) - 1 + \zeta_i\right) = 0 \\ \lambda_i \geq 0 \end{cases}$ for $i = 1, \ldots, n$

(V) $\begin{cases} \gamma_i \zeta_i = 0 \\ \gamma_i \geq 0 \end{cases}$ for $i = 1, \ldots, n$

Non-Separable SVM Remarks

- For all points on the correct side of the decision boundary:

$$\zeta_i = 0 \quad \text{and} \quad y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) > 1 \quad \Rightarrow \quad \lambda_i = 0$$

- The support vectors ($\lambda_i \neq = 0$) are points on the border, inside the margin, and on the wrong side of the decision boundary.

- For all points inside the margin or the wrong side:

$$\zeta_i > 0 \quad \text{and} \quad \lambda_i = C \quad \Rightarrow \quad \gamma_i = 0$$

- For support vectors on the margin border:

$$\zeta_i = 0 \quad \text{and} \quad 0 \leq \lambda_i \leq C \quad \Rightarrow \quad \gamma_i \geq 0$$

Unconstraint Optimization for SVM (Non-Separable Case)

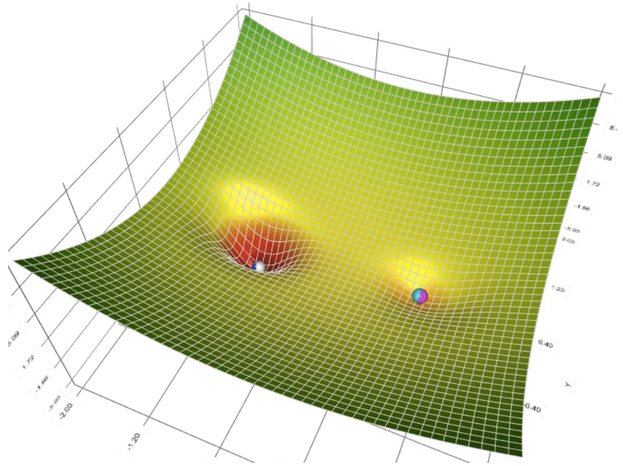
The dual problem

The optimization now can be written as a function of *the Lagrange multiplier*, $\boldsymbol{\lambda}$

$$\max_{\lambda} \quad \sum_{i=1}^{n} \lambda_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j}^{n} \lambda_i \lambda_j \, y_i y_j \, \boldsymbol{x}_i^\top \boldsymbol{x}_j$$

$$\text{s.t.} \quad 0 \leq \lambda_i \leq C, \qquad \text{for } i = 1, \ldots, n$$

$$\sum_{i=1}^{n} \lambda_i y_i = 0$$



*Now, one may do variants of **gradient descent** to find $\boldsymbol{\lambda}$ minimizing the loss. This is topic of future lectures.*

Takeaways

An optimization perspective toward machine learning methods (think visually)

scikit learn for SVM

1.4. Support Vector Machines

Support vector machines (SVMs) are a set of supervised learning methods used for classification, regression and outliers detection.

The advantages of support vector machines are:

- Effective in high dimensional spaces.
- Still effective in cases where number of dimensions is greater than the number of samples.
- Uses a subset of training points in the decision function (called support vectors), so it is also memory efficient.
- Versatile: different Kernel functions can be specified for the decision function. Common kernels are provided, but it is also possible to specify custom kernels.

The disadvantages of support vector machines include:

- If the number of features is much greater than the number of samples, avoid over-fitting in choosing Kernel functions and regularization term is crucial.
- SVMs do not directly provide probability estimates, these are calculated using an expensive five-fold cross-validation (see Scores and probabilities, below).

https://scikit-learn.org/stable/modules/svm.html